BioRAG-DRAG: Multimodal Sequence-to-Literature Evidence for Local-First Biomedical Agents

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Abstract

Biomedical agents need an auditable route from heterogeneous queries to biological evidence, yet ordinary text retrieval-augmented generation does not preserve the alignment semantics of DNA or protein search, while BLAST does not provide a shared context interface. We present **BioRAG-DRAG**, a local-first multimodal retrieval layer that combines partition-specific neural encoders, BLAST verification, and typed graph evidence. BioRAG-Standard v0 exports 257,886 text, DNA/cDNA, protein, and mixed records with an initial engineering annotation layer. The architecture treats vector retrieval as a coarse candidate and context route rather than a replacement for alignment search.

On an in-index sequence-window stress test, vector retrieval recovers substantial but lower biological match than BLAST. Held-out parent controls select ProtT5 for protein partitions and DNABERT-2 for the auxiliary DNA candidate route, while BLAST remains stronger for verification. We then build a 2,000-protein SeqLit-DAG linking sequence candidates to GO/GOA and PubMed evidence. Two 100-query cluster controls remove either complete observed UniRef50 groups or full identity-30 BLASTP components. ProtT5 improves function-linked candidate ranking over BLASTP in both controls, whereas simple reciprocal-rank fusion is not robust. On an accession-masked 33/66 development/test split, a fixed Qwen3.5-9B executor isolates ordinary text-RAG, sequence-vector, combined BLAST+vector, and typed DAG evidence. The live text-RAG control does not place expected evidence in the fixed raw-sequence prompt, while sequence-aware retrieval gives a resolved gain. BLAST fusion directionally increases prompt-level evidence coverage; DAG compression does not change that coverage, but removes a significant fraction of GO evidence-selection errors under the same candidate budget. Across evidence-bearing routes, every emitted citation is entailed by the supplied pack, no GO/PMID identifier is introduced from outside the pack, and mechanism questions correctly abstain. These structured tasks do not establish free-form biomedical reasoning or mechanism discovery; they demonstrate a reproducible sequence-to-literature evidence interface for local agents.

1 Introduction

Biomedical information retrieval is inherently multimodal. A single user question may involve a gene symbol, a DNA fragment, a protein sequence, a pathway description, a variant, a paper abstract, or a combination of these. Traditional bioinformatics systems expose these modalities through separate tools: BLAST for local alignment [1], MMseqs2 or DIAMOND for high-throughput sequence search [2, 3], SQL or keyword search for database records, ontology lookup for GO terms, pathway databases for molecular processes, and graph traversal for relationships among entities. This division is scientifically useful, but it creates a practical gap for large language model (LLM) agents. Agents need a unified evidence interface that can retrieve, cite, and reason over heterogeneous local data without forcing the user or the model to manually choose a tool for each modality.

Retrieval-augmented generation (RAG) offers a natural interface for evidence grounding [4], but ordinary text RAG is not sufficient for biological data. DNA and protein sequences are not just strings of natural language, and exact sequence similarity has established statistical and biological foundations. Therefore, a biological RAG system should not frame neural vector retrieval as a replacement for BLAST. Instead, vector retrieval should act as a fast, shared candidate layer, while classical tools provide verification where their assumptions fit the query.

We propose **BioRAG-DRAG**, a multimodal biological retrieval layer for local-first biomedical agents. The core design is:

query / sequence / mixed input
→ partitioned vector retrieval for coarse candidates and rapid context
→ BLAST verification or reranking for sequence-level evidence
→ DRAG graph expansion for typed evidence paths
→ local-first agent answer with citations and retrieval traces

This design separates three roles. First, vector retrieval provides a shared interface over text, DNA/cDNA, protein, and mixed records, while allowing different partitions to use different embedding models. Second, BLAST remains the biological alignment route for verification and reranking. Third, DRAG turns retrieval results into graph evidence that an agent can inspect, cite, and expand. The system is therefore complementary to BLAST rather than competitive with it.

The current empirical result supports this division as an engineering validation rather than a held-out homology benchmark. On the BioRAG-Standard v0 sequence-window stress test, vector retrieval is not as strong as full-database BLAST for alignment-grounded sequence verification, but it often places a biologically matched parent within a 200-candidate pool for downstream alignment-based verification. Held-out parent-fragment controls further show that protein-specialized ESM-2 and ProtT5 embeddings are stronger than the current OmniGene protein-window embedding, while a DNA-specific DNABERT-2 mean-pooling encoder provides a better candidate layer than the unified DNA control. The DNA route remains below BLASTN. A separate SeqLit-DAG experiment tests the application-level claim: whether a raw protein sequence can lead to compact, typed function and paper evidence that a local instruction model can cite without leaving the retrieved pack. Accession-masked, observed-UniRef50-cluster, and identity-30 component controls progressively tighten leakage while preserving this application definition.

This paper makes four contributions:

1. **A reproducible local corpus/library export.** We build BioRAG-Standard v0 from Open-Rosalind Standard data and sequence-window extensions, yielding 257,886 corpus records and an initial annotation layer for engineering evaluation.
2. **A local-first hybrid retrieval architecture.** We implement vector coarse retrieval, BLAST verification, and DRAG graph packaging over Standard biomedical text, protein windows, DNA/cDNA windows, and mixed records.
3. **A retrieval and engineering evaluation.** We compare BLAST, vector retrieval, vector reranking, combined vector+BLAST evidence routes, held-out protein embedding baselines with ESM-2 and ProtT5, and an initial public DNA encoder control; we also decompose indexed lookup and verified-route latency.
4. **A sequence-to-literature Agent application.** We release a typed protein-to-GO/GOA-to-PubMed DAG view, add observed-UniRef50 and identity-30 cluster-held-out controls, compare a live ordinary text-RAG

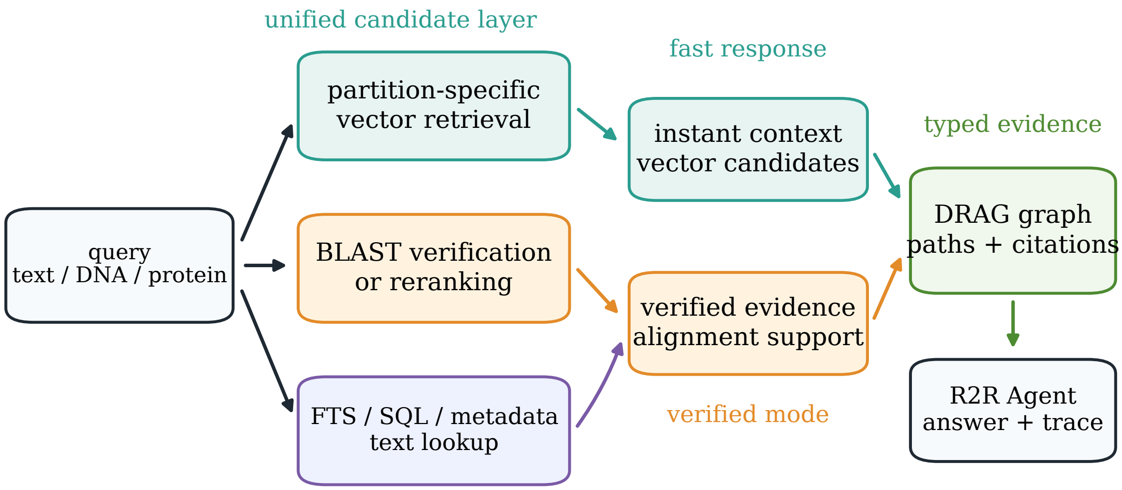

Figure 1: BioRAG-DRAG architecture. Vector retrieval provides a rapid multimodal candidate layer after query embedding; BLAST supplies sequence verification and reranking; DRAG packages vector, alignment, text, and graph evidence into typed traces for a local biomedical agent.

route with sequence-aware evidence routes under a fixed executor, and retain biological graph analysis as an explicitly exploratory result bounded by parent-collapse, k-mer, and degree-preserving null controls.

2 Related Work

Classical sequence search. BLAST remains a central tool for sequence similarity because it provides alignment-based scoring and interpretable statistical evidence [1]. Later systems such as MMseqs2 and DIAMOND improve speed and sensitivity at large scale, especially for protein and metagenomic workloads [2, 3]. This line of work is not a weak baseline for BioRAG; it is the biological verification layer that BioRAG should preserve. Our question is different: whether a neural multimodal retriever can provide a fast, shared candidate and context layer that cooperates with alignment search rather than replacing it.

Biomedical language models and biomedical RAG. Domain-specific biomedical language models such as BioBERT, PubMedBERT, BioGPT, and BioMedLM improve biomedical text representation or generation by pretraining on biomedical corpora [5, 6, 7, 8]. RAG systems then combine model generation with retrieved evidence to reduce reliance on parametric memory [4]. Clinical and biomedical RAG systems such as Almanac and recent medical GraphRAG variants show the value of retrieval and graph evidence for safer medical question answering [9, 10]. BioRAG-DRAG is complementary to these systems: it focuses on local-first biological retrieval across text, DNA/cDNA, protein windows, and graph evidence, rather than on clinical QA alone.

Biological foundation models and learned sequence search. Protein language models show that sequence-only pretraining can learn useful structure and function representations [11, 12, 13]. Dedicated protein-retrieval work such as PLMSearch demonstrates that learned protein representations can support fast remote-homology search [14]. More recently, ERAST combined large language models and vector database technology for scalable homology detection over approximately one billion biological sequences, covering both nucleotide and protein sequences [15]. Genomic language models, including DNABERT, DNABERT-2, Nucleotide Transformer, HyenaDNA, and Evo, similarly show that DNA-scale pretraining can produce transferable sequence representations [16, 17, 18, 19, 20]. These systems make clear that dense biological sequence search is not new by itself. BioRAG-DRAG instead studies how dense sequence-text retrieval can be packaged as a local biomedical-agent evidence layer together with BLAST verification, text records, and graph traces. We treat the embedding layer as pluggable: protein-only partitions can use public encoders such as ESM-2 or ProtT5, while OmniGene-4 CPT serves as a unified biological-language backbone configured with additional DNA, protein, and structure-token vocabulary entries. An activation audit reported below prevents attributing the current retrieval results to those added entries. Answer generation remains a separate agent/model function.

Vector databases and graph-augmented retrieval. Dense vector retrieval has become a common substrate for RAG because approximate nearest-neighbor indexes can serve large precomputed embedding collections efficiently; FAISS is a representative GPU-capable system for billion-scale similarity search [21]. Graph-augmented RAG extends this substrate by organizing retrieved records and relations as paths or communities rather than isolated chunks; GraphRAG is one prominent recent example for text corpora [22]. Biological data are naturally graph structured: retrieved sequences often need to connect to genes, proteins, annotations, pathways, variants, and literature. BioRAG-DRAG evaluates both method-agnostic vector-neighbor graphs and BLAST-enriched hybrid graphs, asking whether biological structure appears before explicit domain rules are added.

Local biomedical agents and this work's lineage. BioRAG-DRAG builds on two local prerequisites. Open-Rosalind defines the auditable biomedical-agent contract: tool-mediated evidence, trace completeness, and bounded workflows [23]. OmniGene-4 provides a Gemma-derived biological sequence-text CPT checkpoint whose tokenizer contains additional DNA, protein, and structure-token entries [24]. This paper connects those directions by turning biological representations into a retrieval substrate and by packaging retrieval results into the evidence traces expected by a local biomedical agent; the retrieval layer itself remains model-agnostic.

3 BioRAG-Standard v0 Corpus

3.1 Corpus Construction

BioRAG-Standard v0 is built from the existing Open-Rosalind Standard index and local sequence-vector extensions. The exported dataset is stored under `data/biorag_standard_v0` and uses JSONL files for both corpus records and annotated tasks.

Each corpus row contains a stable dataset-local ID, source record ID, biological entity ID, source accession, modality, partition, retrievable text or sequence, extracted biological labels, and traceable metadata. Sequence-window records preserve parent accession, parent record ID, window offset, window size, stride, alphabet, and source FASTA header.

Table 1: BioRAG-Standard v0 corpus partitions.

Corpus partition	Records	Source
Standard text	57,856	HGNC genes, GO terms, Reactome pathways, ClinVar gene summaries
Protein sequence windows	100,000	Swiss-Prot windows embedded with OmniGene CPT
DNA/cDNA sequence windows	100,000	Ensembl cDNA windows embedded with OmniGene CPT
Mixed POC records	30	mixed English/sequence examples
Total	**257,886**	text + protein + DNA/cDNA + mixed

Table 2: BioRAG-Standard v0 annotation families.

Task family	Records
Gene lookup	5
Pathway lookup	3
Protein sequence retrieval	52
DNA/cDNA sequence retrieval	52
Total	**112**

3.2 Annotated Tasks

The current annotation layer contains 112 retrieval tasks:

All tasks have at least one `positive_corpus_refs` link in the full export, enabling evaluation by direct corpus references as well as by expected labels such as entity IDs, source IDs, accessions, symbols, and biological gene labels.

3.3 Role in the Study

The corpus/library is the shared substrate for all retrieval conditions:

- text lookup over Standard records,
- vector search over Standard text and sequence windows,
- BLAST over sequence records,
- combined BioRAG retrieval with BLAST, vector, text, and graph routes,
- DRAG graph construction and biological enrichment analysis.

This shared substrate makes ablations cleaner: removing DNA, protein, text, graph, or BLAST routes changes the retrieval view without changing the underlying evidence universe. We do not claim that this v0 task layer is a comprehensive biological retrieval benchmark; it is an exported local evidence library with enough annotations to validate the first BioRAG workflow.

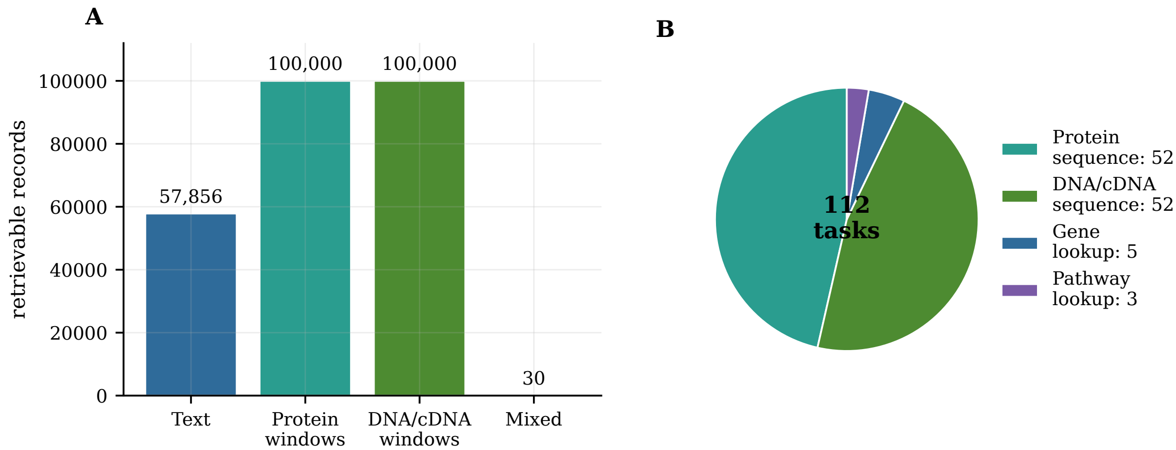

Figure 2: BioRAG-Standard v0 composition. The corpus/library contains 257,886 retrievable records across Standard text, protein windows, DNA/cDNA windows, and mixed records, with a first 112-task annotation layer.

4 Method

4.1 Typed Multimodal Records

Records are typed before embedding or indexing. Examples include Standard biomedical text, protein sequence windows, DNA/cDNA sequence windows, and mixed English/sequence records. For sequence-window retrieval, the document body is the biological sequence itself, while metadata carries the parent accession and source information. This avoids embedding long header-heavy FASTA records as single documents and improves fragment retrieval.

The embedding layer is partitioned and pluggable. The main BioRAG-Standard sequence-window stress test uses the full merged OmniGene CPT model in BF16 so that DNA, protein, mixed sequence/text, and agent-facing biological-language inputs share one representation backbone. For protein-only held-out controls, we also build public ESM-2 and ProtT5 sequence-window indexes; for DNA/cDNA, we compare Nucleotide Transformer 500M, DNABERT-S, and DNABERT-2. Embeddings are computed from the final hidden layer using attention-mask mean pooling over token hidden states followed by L2 normalization, not from next-token prediction logits. We also run last-token pooling, CLS-token pooling, and reverse-complement averaging where supported as ablations. The main controlled DNA result uses DNABERT-2 mean pooling with 256-nucleotide windows and stride 128; 128/64 is retained as a window ablation. The Chroma collections contain 100,000 DNA/cDNA windows and 100,000 protein windows for the main stress test; the controlled20k held-out protein control expands to 112,808 protein windows, and the controlled100k ProtT5 protein control expands to 589,210 windows. A tokenizer audit found that the checkpoint contains 19,994 marker-prefixed DNA and 7,974 marker-prefixed protein BPE entries, but plain raw sequences activate none of them; samples of ten million integers from each DNA and protein CPT binary part likewise contained no such IDs. The OmniGene retrieval rows therefore evaluate sequence CPT through the base tokenizer and do not isolate a vocabulary-expansion effect.

4.2 Vector Coarse Retrieval

Vector retrieval serves two purposes:

1. **Instant mode.** It returns fast provisional context for interactive use.

2. **Candidate generation.** It produces top-N sequence/entity candidates for downstream biological verification.

For sequence-window queries, long inputs are split into query windows, then results are merged across windows. A lightweight sequence-aware reranker can optionally combine vector similarity with sequence overlap features. This reranker is not intended to replace BLAST; it is used to test whether simple local features can improve the vector candidate layer.

4.3 BLAST Verification and Reranking

For sequence inputs, BLASTP or BLASTN is used as the alignment-grounded verification route. In the current implementation, BLAST is run against local Swiss-Prot and Ensembl cDNA databases. The intended pipeline is:

```
vector top-N candidates
  -> BLAST fine scoring or full-database fallback
  -> reranked evidence with alignment metadata
```

We evaluate both full-database BLAST and candidate-subset BLAST. In the candidate-subset condition, vector retrieval first selects top-N parent accessions, `blastdbcmd` extracts those candidates from the local BLAST database, and BLASTP/BLASTN reranks only that candidate FASTA. This condition measures whether vector retrieval can supply a useful candidate set for verified BioRAG, while full-database BLAST remains the reference verification route.

4.4 DRAG Evidence Graphs

DRAG graphs convert retrieved records and relations into graph evidence. We build three sequence-window graph views:

- **vector-only graph:** edges are nearest-neighbor vector relations;
- **BLAST-only graph:** edges are alignment-derived sequence neighbors;
- **hybrid graph:** vector and BLAST edges are merged while preserving edge types.

The vector-only graph deliberately follows a method-agnostic text-RAG recipe: records become nodes, nearest-neighbor retrieval creates edges, and biological labels are used only for downstream analysis. This lets us test whether biological structure emerges before explicit biological rules are added.

4.5 Sequence-to-Literature DAG and Agent Contract

The application DAG uses curated identifiers rather than extracting claims from paper prose. Its typed path is

$$q_{\mathrm{seq}} \to p_{\mathrm{candidate}} \to g_{\mathrm{GO/GOA}} \to d_{\mathrm{PMID}} \to a_{\mathrm{agent}}.$$

The 2k resource contains 2,000 Swiss-Prot proteins, 12,858 GO annotations, 4,304 PMIDs, and 63,651 typed edges. From this resource we construct a function-to-paper split with 99 held-out protein accessions and 1,901 index proteins. Held-out accessions and full-sequence substrings have zero index overlap. Relevance is defined by low-frequency shared GO terms with index-side curated GOA paper evidence, so the task measures recovery of traceable annotation and literature paths, not direct experimental validation of a new query protein.

We add two 100-query sequence-cluster controls over the same frozen resource. The first removes every protein assigned to the query's observed UniRef50 cluster, leaving 1,892 index proteins and excluding eight non-query cluster members. Because 1,969 of 1,984 observed clusters are singletons, this is a same-UniRef50-cluster leakage control rather than a strong family-diversity sample. The second builds BLASTP connected components at at least 30% pair identity and 80% coverage of the shorter sequence, selects queries only from non-singleton components, and removes each selected component in full. It leaves 1,732 index proteins and excludes 168 non-query proteins. Both audits find no cluster, exact accession, exact 160-residue query-fragment substring, or full-sequence substring overlap; identity-30 additionally has no threshold-qualifying cross-split pair. This identity-30 split is deliberately cluster-stratified and is reported as a stress test, not a prevalence-weighted or Pfam-clan-held-out benchmark.

For executable evidence selection, Graph-IDF scores a claim using candidate rank, corpus GO document frequency, and typed GO-to-PMID support. Hyperparameters are selected on a deterministic 33-query development split and frozen on 66 test queries for the accession-masked application experiment. The cluster controls repeat this protocol with 33 development and 67 frozen test queries per route; test labels are never exposed to the selector. The accession-masked selected configuration considers five candidates, uses IDF power 1.5, and emits at most three GO claims or five PMID claims. Qwen3.5-9B [26] is the main fixed executor, with Qwen2.5-7B-Instruct [25] retained as a generator-robustness control; neither model is used as the biological retrieval encoder. Prompts mask the held-out accession and expose only typed evidence identifiers. Function and literature questions require copied GO or PMID identifiers with citations; mechanism questions require explicit abstention because no mechanism-level evidence is supplied.

To operationalize the remaining free-form evaluation gap, we additionally freeze an assessor-blinded pilot before collecting any ratings. Thirty of the 66 test queries are sampled without reference to retrieval outcomes, with ten queries each from single-GO/sparse-literature, single-GO/dense-literature, and multi-GO strata. One fixed Qwen3.5 executor generates a four-part scientific evidence note for each of five routes: no retrieval, ordinary R2R text retrieval, sequence vector retrieval, combined BLAST+vector evidence, and combined evidence with DAG compression. Each normalized prompt exposes at most five candidates, three GO claims, and three PubMed records. A seed-randomized balanced Latin square masks route labels across variants A–E. The frozen protocol specifies two independent domain reviewers and a primary utility endpoint averaging functional correctness, citation support, and calibration. Because human ratings have not yet been collected, this pilot is not used as evidence for free-form biomedical reasoning in the present results.

Every SeqLit-DAG relationship records its predicate, source database, supporting source record, evidence level, optional retrieval score, verification method, and database snapshot. Curated protein–GO–paper edges retain the GOA evidence record and evidence code; query-time vector and BLAST edges remain distinguishable as unverified dense retrieval and alignment verification. For the application projection, R2R v3 [27] provides document CRUD, collections, access control, ordinary text retrieval, and Agent APIs. BioRAG supplies partition-specific sequence retrieval, BLASTP/BLASTN verification, and typed evidence packs. The authoritative projection imports explicit entities and relationships rather than asking an LLM to re-extract biological claims from prose, and raw sequences are excluded from its generic text index by default. For the controlled ordinary text-RAG ablation, a separate leakage-audited R2R collection includes 1,901 index-protein records and 12,170 flattened evidence records, with zero exact overlap against the 99 held-out parent accessions. It uses Qwen3-Embedding-0.6B [28] through Ollama (Q8_0, 1,024 dimensions); graph, full-text, and hybrid search are disabled so that only generic semantic text retrieval changes.

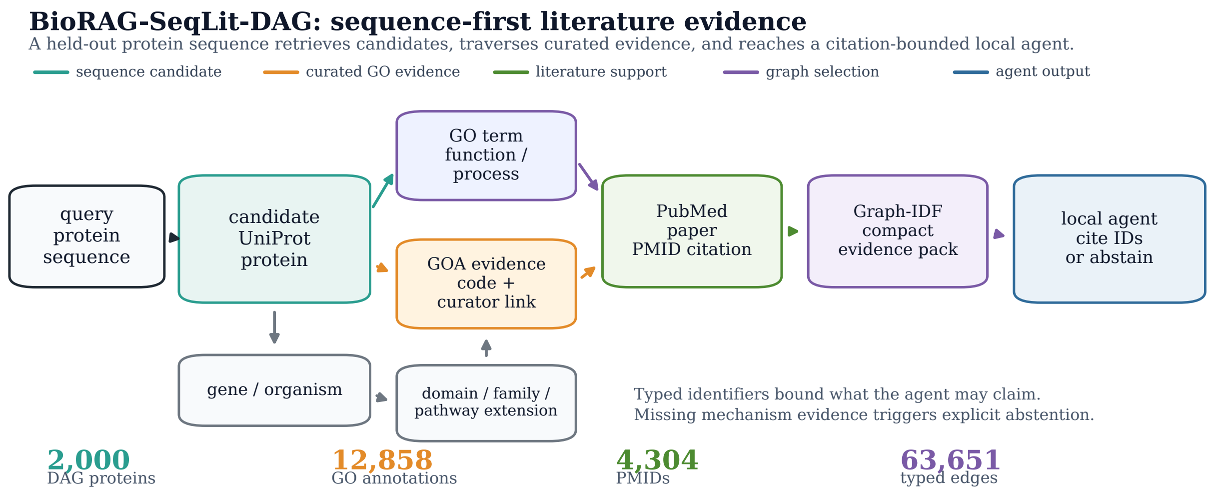

Figure 3: BioRAG-SeqLit-DAG application path. A held-out protein sequence retrieves candidate proteins, traverses curated GO/GOA-to-PubMed relations, and is compressed by a Graph-IDF selector into a citation-bounded Agent evidence pack. Typed identifiers limit valid claims, while absent mechanism evidence triggers abstention.

5 Experiments

5.1 Retrieval Conditions

We evaluate the current local implementation as a workflow validation and stress test:

- **BLAST:** local BLASTP/BLASTN sequence retrieval;
- **Vector:** sequence-window Chroma retrieval using the configured embedding model;
- **Vector + sequence rerank:** vector retrieval plus a lightweight heuristic sequence-overlap reranker;
- **Combined BLAST+vector evidence:** BLAST and vector routes are both executed for sequence queries, with FTS, graph, and vector routes available for text and mixed records.

Metrics include Hit@k, MRR, biological Hit@k, biological MRR, average latency, route coverage, and retrieval traces.

The 100-query sequence-window set is not a held-out homology benchmark: queries are fragments or perturbations derived from indexed parent sequences. We therefore use it to test local retrieval behavior, candidate-pool quality, and evidence packaging. Separately, we evaluate a 500-query held-out parent-fragment control over a controlled20k protein index to compare BLAST, OmniGene, and public ESM-2 embeddings under a more realistic no-exact-parent setting. The SeqLit-DAG application then uses the accession-masked split and the two cluster controls above. The cluster experiments compare random candidates, fixed 3-mer Jaccard, BLASTP, ProtT5 mean embeddings, and a saved-ranking ProtT5+BLAST reciprocal-rank fusion ablation. Query-level paired bootstrap intervals use 10,000 resamples.

5.2 Engineering Latency

Latency is decomposed into vector-index lookup, BLAST route time, graph expansion, verified-mode median sum, and FAISS CPU lookup. All vector lookup measurements exclude model cold start, query embedding

Table 3: Retrieval quality on the 100-query BioRAG-Standard sequence-window stress test. This evaluates local fragment recovery and candidate-pool behavior, not held-out biological homology search.

Condition	Exact Hit@10	Exact MRR	Biological Hit@10	Biological MRR	Scope
BLAST	0.73	0.5425	0.9899	0.9768	full local BLAST database
Vector	0.50	0.4063	0.8182	0.7088	OmniGene/Chroma
Vector + sequence rerank	0.70	0.5987	0.9293	0.9217	vector plus heuristic sequence-overlap features
Combined BLAST+vector evidence	0.73	0.5516	0.9899	0.9775	BLAST plus vector/graph traces

generation, LLM generation, and network calls. This isolates the indexed retrieval stage after a resident embedding service has produced the query vector.

5.3 Graph Evidence Structure

We evaluate whether DRAG can package retrieved sequence records into typed, inspectable graph neighborhoods. The main text reports a qualitative evidence-trace visualization and a compact graph-control summary. Community purity, GO/Reactome enrichment, and PubMed-sharing analyses are treated as exploratory appendix results because they remain underpowered for biological mechanism claims. An auxiliary parent-level token graph compares the intended OmniGene protein BPE segmentation, the actual runtime tokenizer, and overlapping fixed 3-mers. Token–GO associations use hypergeometric tests with Benjamini–Hochberg correction over all eligible pairs; complete GO label sets are permuted within protein-length strata as a null control. For the retrieval ablation, token–GO edges are learned from 1,901 index proteins only. A 33-query development split selects the graph-expansion weight and the number of ProtT5 candidates preserved before token-based tail replacement; direct token BM25, token–GO expansion, fixed 3-mer BM25, reciprocal-rank fusion, and the frozen tail route are then evaluated on the same 66 held-out queries used by the Agent experiment.

6 Results

6.1 Retrieval Quality

On the 100-query BioRAG-Standard sequence-window stress test, the retrieval results are shown in Table 3.

BLAST remains the strongest verification route. Vector-only retrieval is weaker on exact parent-ID recovery, but it recovers biologically related neighborhoods often enough to serve as a useful coarse-retrieval layer in this local setting. The heuristic sequence-aware reranker narrows much of the gap; we treat this as an engineering ablation rather than a methodological contribution. The combined BLAST+vector evidence route preserves BLAST-level biological recall while retaining vector neighborhoods and route traces for agent use. These results should not be interpreted as evidence of held-out biological retrieval competitiveness.

As a candidate-pool ablation, vector-to-candidate-BLAST reranking reaches biological Hit@10/MRR of 0.9293/0.9293 with 200 vector-selected parent candidates, with candidate biological Recall@200 of 0.9596. This shows that, in this in-index local stress test, vector retrieval often places the biologically matched parent within a 200-candidate pool for downstream alignment-based verification. It does not establish a speed advantage; matched 100k/300k/full vector scale curves are required before candidate-BLAST can be claimed as a systems improvement. The full budget sweep is therefore reported only in Appendix A.

The interpretation is therefore not "vector beats BLAST." The result supports a two-stage engineering strategy: vector retrieval supplies unified candidates and provisional context; BLAST supplies alignment-grounded verification.

Table 4: Held-out parent-fragment protein retrieval on the controlled20k index. Vector rows use sequence-window Chroma retrieval with a top-200 candidate pool.

Method	Pooling	Biological Hit@10	Biological MRR	Recall@200	Mean latency
BLAST	alignment	0.8560	0.8289	0.8560	138.6 ms
OmniGene-4-CPT BF16	mean	0.3120	0.2623	0.4940	538.4 ms
OmniGene-4-CPT BF16	last-token	0.0940	0.0725	0.1840	531.5 ms
ESM-2 650M	mean	0.6560	0.5764	0.7340	329.0 ms
ESM-2 650M	last-token	0.2360	0.2025	0.3560	332.2 ms
ProtT5-XL-UniRef50	mean	0.7780	0.7158	0.8460	329.1 ms
ProtT5-XL-UniRef50	last-token	0.4680	0.3668	0.6400	333.1 ms

6.2 Held-Out Protein Embedding Baselines

To separate BioRAG system design from the choice of embedding backbone, we also run a 500-query held-out parent-fragment control on a controlled20k protein index. The exact held-out parent accessions are excluded from the index; the main metric is whether retrieval finds an indexed biologically related record. Table 4 shows that BLAST remains the strongest alignment-grounded route, while public protein encoders are substantially stronger than the current OmniGene protein-window embedding on this protein-only retrieval task. ProtT5 mean pooling is the strongest completed dense baseline. Last-token pooling is weak across OmniGene, ESM-2, and ProtT5, so mean pooling remains the default embedding mode.

This result changes the interpretation of OmniGene in BioRAG. OmniGene should not be presented as the strongest protein-only dense retriever in the current experiments, and last-token pooling is a poor choice for this fragment-retrieval setting. Its role is better framed as a unified biological-language backbone for mixed sequence/text and agent workflows. BioRAG can use specialized encoders such as ProtT5 or ESM-2 for protein partitions while keeping the same Chroma, BLAST, DRAG, and citation interfaces.

6.3 Held-Out DNA/cDNA Control

We also construct a 100-query DNA/cDNA parent-fragment control from Ensembl cDNA. Exact held-out transcript accessions are removed from the index, and biological matching uses shared `gene_symbol` labels. To make the small vector-index experiment interpretable, we build a controlled20k cDNA subset containing at least one non-held-out same-gene transcript for every query plus random background records. This split has zero exact held-out transcript leakage, but 62 of 100 queries are exact substrings of other indexed transcripts, reflecting isoform and conserved-fragment overlap; we therefore treat it as a held-out transcript-fragment control rather than a strict remote-homology benchmark.

This DNA result is intentionally reported because it prevents overclaiming. DNABERT-2 mean pooling with 256/128 windows is the strongest tested dense DNA route, but remains well below BLASTN on the alignment-grounded reference. A scratch-trained DNA-ESM2-style 117M MLM encoder reaches 0.2300/0.1687 Bio Hit@10/MRR, close to but below DNABERT-2, so simply increasing model size or using an ESM-style architecture is not sufficient. DNABERT-2 improves over the OmniGene control and the weaker Nucleotide Transformer control, while last-token pooling collapses and reverse-complement averaging does not improve the primary Hit@10 result. We therefore do not conclude that dense DNA retrieval replaces alignment search. The result reinforces the central system design: BioRAG should keep a unified evidence interface while allowing the DNA partition to use retrieval-specific genomic encoders, fine-tuning, or alignment verification.

Table 5: Held-out DNA/cDNA parent-fragment control. BLASTN is the alignment-grounded reference. Vector rows use top-200 parent collapse on the controlled20k window set; latency is reported by the corresponding evaluator and is not a cold-start measurement.

Method	Index setting	Biological Hit@10	Biological MRR	Mean latency
BLASTN	controlled20k cDNA	0.9100	0.9100	94.7 ms
OmniGene-4-CPT BF16 mean	controlled20k windows	0.1000	0.0795	1049.0 ms
Nucleotide Transformer 500M mean	full controlled20k windows	0.0100	0.0021	992.5 ms
DNABERT-S mean	controlled20k windows, 128/64	0.1200	0.1020	12.6 ms
DNABERT-S mean + RC average	controlled20k windows, 128/64	0.1200	0.1090	12.8 ms
DNABERT-2 mean	controlled20k windows, 128/64	0.1700	0.1481	7.4 ms
DNABERT-2 mean	controlled20k windows, 256/128	0.2500	0.2347	10.9 ms
DNABERT-2 last	controlled20k windows, 128/64	0.0200	0.0171	11.9 ms
DNA-ESM2-style 117M MLM mean	controlled20k windows, 256/128	0.2300	0.1687	n/r
BLASTN	full local cDNA	0.9100	0.9100	328.3 ms

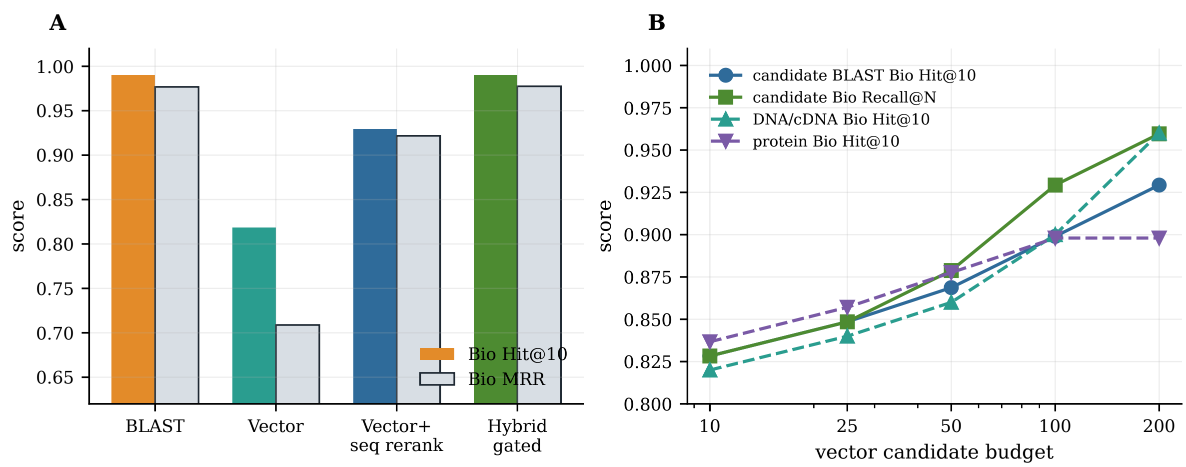

Figure 4: Retrieval quality on the 100-query sequence-window stress test. BLAST remains the strongest alignment-grounded verifier, while vector retrieval supplies lower but useful local candidate context; candidate-BLAST is treated as an ablation rather than a peer condition.

6.4 Engineering Latency

Standard text and 100k BF16 sequence-window indexes give the lookup and verification times in Table 6.

These results support a **lookup/verified** product design. After a query embedding is available from a resident embedding service, vector lookup adds only a small indexed-search overhead; in verified mode, BLAST and DRAG add biological grounding and evidence attribution. We do not report end-to-end instant latency here because query embedding time is excluded. The current Chroma measurements are therefore complemented by a separate controlled DNA backend microbenchmark below.

As a separate controlled DNA backend microbenchmark, we replay 10,000 sampled resident vectors against the 20,000-vector DNABERT-2 mean collection at top-200. Chroma, FAISS CPU, and FAISS GPU require 2.2972, 1.9210, and 0.003868 ms/query, respectively (435, 521, and 258,502 queries/s). These measurements exclude query embedding, host-service overhead, BLASTN, graph expansion, and generation; they establish the serving direction for an instant candidate layer rather than an end-to-end speed claim. The full protocol and JSON artifacts are provided in the supplementary reproducibility report.

Table 6: Indexed lookup and verified-route latency.

Component	Text / Standard	DNA/cDNA	Protein	Notes
Chroma lookup, top-10	4.6924 ms	5.9630 ms	6.0948 ms	query embedding excluded
BLAST top-10	n/a	294.5684 ms	1081.9300 ms	local BLAST
Hybrid graph expansion	n/a	0.3540 ms	0.3254 ms	SQLite graph expansion
Verified vector+BLAST+graph	n/a	300.8221 ms	1088.8706 ms	median route-latency sum
FAISS CPU lookup, IndexFlatIP	8.3346 ms	15.4626 ms	15.6090 ms	flat top-10 lookup
FAISS GPU lookup, DNA controlled20k	n/a	0.003868 ms	n/a	top-200, 10k in-index queries

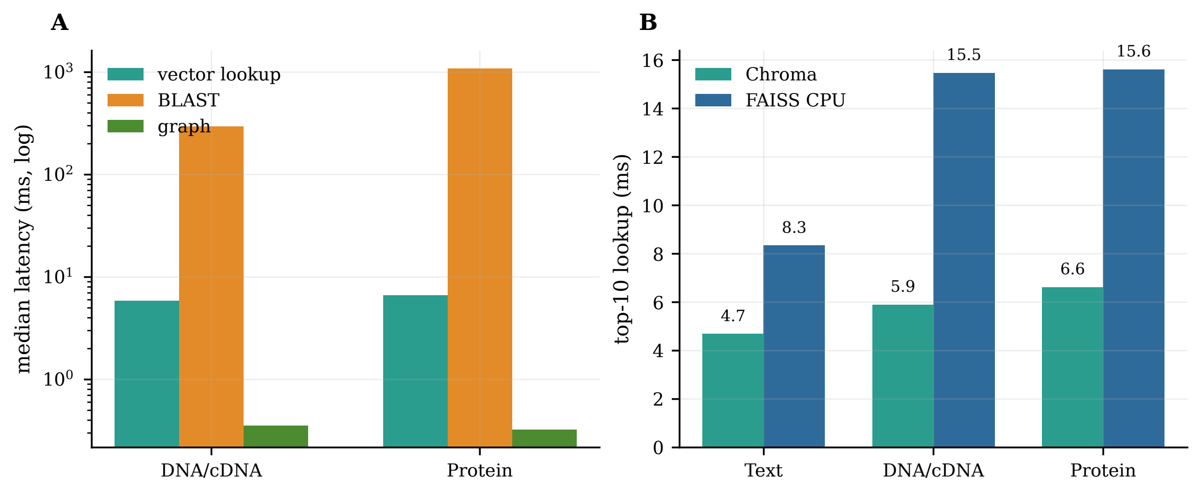

Figure 5: Indexed lookup and verified-route latency. Vector lookup is measured after query embedding; BLAST dominates verified-mode latency, while graph expansion is negligible.

6.5 Swiss-Prot Scale Frontier

We further measure the scale frontier on the held-out protein parent-fragment benchmark. Starting from the full held-out Swiss-Prot-derived index, we create controlled 100k and 300k subsets that preserve same-gene positive candidates for all 500 queries and have zero exact held-out-parent leakage. Table 7 reports completed BLAST references, completed ProtT5 mean vector points at controlled20k, controlled100k, and controlled300k, and vector-to-candidate-BLAST reranking points at controlled100k and controlled300k. The full-scale dense index remains pending.

This table is a boundary result rather than a speed claim. BLAST remains the verified reference. ProtT5 mean vector retrieval provides useful but lower dense retrieval, with biological Recall@200 of 0.8460, 0.8000, and 0.7740 at controlled20k, controlled100k, and controlled300k. Candidate-BLAST improves MRR over vector-only at controlled100k (0.7245 vs. 0.6268) and controlled300k (0.6835 vs. 0.5788), but measured total latency remains slower than full-database BLAST at both matched scales. The candidate-BLAST-only portions are much smaller, 189.2 ms and 209.6 ms after vector candidates are available, so the systems bottleneck in this Chroma POC is vector candidate serving rather than candidate alignment. Thus the current result supports candidate-BLAST as an evidence-quality and routing ablation; a systems speed advantage remains a larger-scale or optimized vector-serving hypothesis.

Table 7: Swiss-Prot protein scale frontier on the 500-query held-out parent-fragment benchmark. Vector rows use ProtT5 mean pooling with top-200 Chroma retrieval. Candidate-BLAST is reported as an ablation, not as a speed advantage in the current Chroma implementation.

Scale	Records	BLAST Bio@10	BLAST MRR	BLAST ms	Vector Bio@10	Vector MRR	Vector R@200	Vector ms	Cand. Bio@10/MRR	Cand. ms
controlled20k	20,000	0.8560	0.8289	138.6	0.7780	0.7158	0.8460	329.1	–	–
controlled100k	100,000	0.8320	0.7960	329.2	0.7260	0.6268	0.8000	581.4	0.7580/0.7245	839.7
controlled300k	300,000	0.8140	0.7613	796.5	0.6760	0.5788	0.7740	812.5	0.7320/0.6835	1020.6
full held-out Swiss-Prot	483,581	0.8060	0.7442	1039.3	–	–	–	–	pending	–

Table 8: Compact DRAG graph controls on 10k graph views. Existing DRAG views are already parent-collapsed. Observed graphs show label-enriched communities and degree-preserving null graphs do not, but k-mer/Jaccard graphs also recover strong sequence-family signals; therefore the biological interpretation remains exploratory.

Modality	Condition	Nodes	Edges	Communities	Top label signal
DNA/cDNA	observed vector graph	579	19,344	9	biotype:IG_V_gene, 21 nodes, purity 1.00
DNA/cDNA	k-mer/Jaccard graph	579	2,111	26	gene family:IGKV, 29 nodes, purity 1.00
DNA/cDNA	degree-preserving null	579	19,344	4.2	no comparable top signal
Protein	observed vector graph	2,623	31,794	16	gene symbol:nbaC, 7 nodes, purity 0.50
Protein	k-mer/Jaccard graph	2,623	9,398	107	gene symbol:pgl, 28 nodes, purity 1.00
Protein	degree-preserving null	2,623	31,794	12.6	no comparable top signal

6.6 DRAG Evidence Graph Showcase

Figure 6 shows two real DRAG traces. The DNA/cDNA example centers on an IGKV sequence neighborhood, while the protein example centers on a `yfbR` neighborhood. In both cases, the graph view exposes which edges come from vector similarity and which come from BLAST-derived alignment evidence. This is the main agent-facing value of DRAG in the current paper: retrieved biological sequences are not only ranked, but packaged as typed neighborhoods that can be inspected, cited, and expanded.

These examples are exploratory. Appendix B reports community purity, GO/Reactome enrichment, and shared-PubMed analyses. The controls in Table 8 support non-random graph structure, but they also show that simple sequence similarity explains part of the signal.

6.7 Agent Evidence Case Study

To test the agent-facing evidence contract without adding another generator model, we convert existing DRAG traces into answer-ready evidence packs. No LLM is called in this experiment. The case-study script summarizes the graph context that a downstream agent would receive, including route labels, representative edges, and an explicit boundary on what may be claimed from the available evidence.

Table 9 shows the operational distinction between retrieval routes. For the `IGKV2-40` case, the hybrid pack contains BLAST-supported cDNA neighbors with 100% identity as well as high-score vector neighbors from the same immunoglobulin family. For the protein `pgl` and `yfbR` cases, vector traces recover broad same-label neighborhoods, while BLAST edges provide alignment-labeled support for citation. The resulting answer skeleton is deliberately conservative: the agent may describe the retrieved neighborhood and cite whether each edge is vector-derived or BLAST-derived, but it should not infer function, mechanism, clinical relevance, or experimental conclusions unless annotation, pathway, domain, or literature evidence is also retrieved.

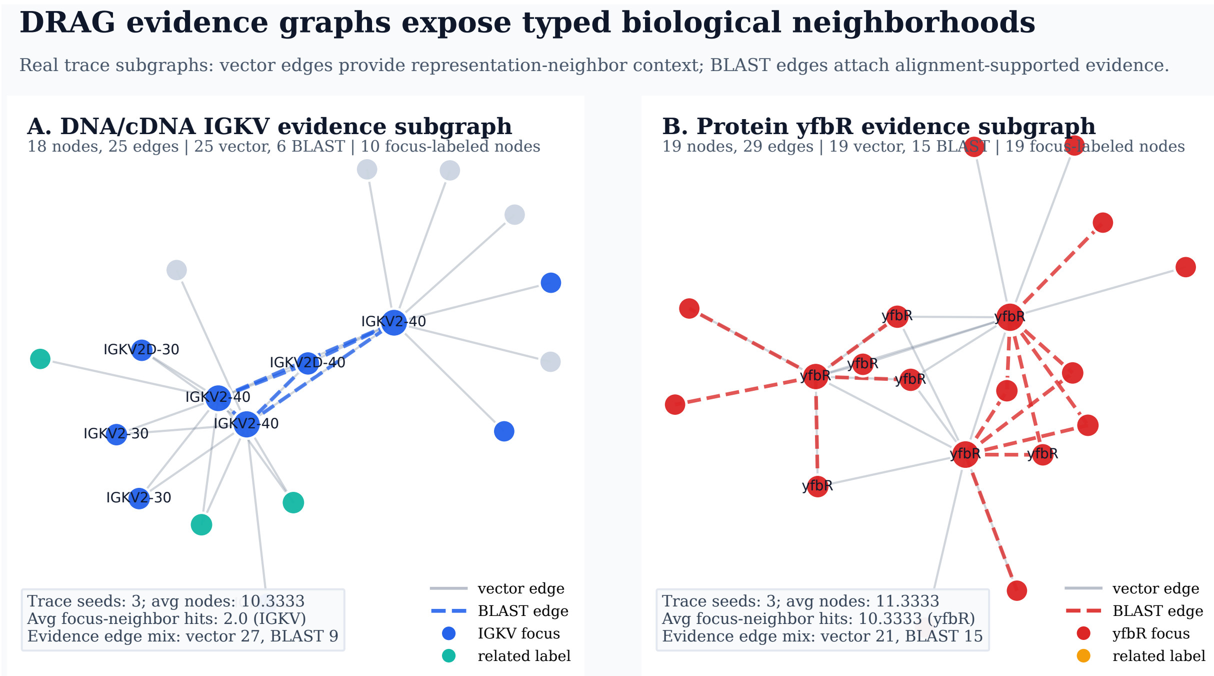

Figure 6: DRAG evidence graph showcase from real local traces. Panel A shows an IGKV-focused DNA/cDNA subgraph; panel B shows a `yfbR`-focused protein subgraph. Solid gray edges are vector-neighbor evidence and dashed colored edges are BLAST-neighbor evidence. The figure illustrates the agent-facing value of DRAG: retrieved biological sequences are packaged as typed, inspectable neighborhoods rather than isolated ranked hits.

Table 9: Agent-style evidence packs generated from DRAG traces. Hybrid DRAG keeps both broad vector context and BLAST-labeled verification edges visible to the downstream agent.

Focus label	Modality	Avg nodes	Avg edges	Avg same-label hits	Relation evidence in hybrid pack
IGKV2-40	DNA/cDNA	10.3	12.0	2.0	BLAST 9; vector 27
pgl	Protein	10.7	12.0	9.7	BLAST 20; vector 16
yfbR	Protein	11.3	12.0	10.3	BLAST 15; vector 21

6.8 Held-Out Agent Evidence Evaluation

The case studies above demonstrate the evidence format, but do not quantify held-out behavior. We first evaluate the deterministic evidence-pack contract on all 99 sequence-to-function-to-literature queries. For each ranked result, the evaluator materializes at most 50 protein candidates and 20 papers, together with GO annotations and typed protein-to-GO-to-PMID paths. Table 10 uses full stored top-50 rankings for every method; this avoids giving methods different effective candidate budgets.

ProtT5 gives the strongest candidate and GO coverage. Reciprocal-rank fusion with BLAST lowers candidate hit by 0.020 but raises the strict typed-path rate by 0.010, so fusion is complementary at the citation path rather than uniformly superior. BLAST alone remains competitive for paper support despite lower candidate coverage.

Table 11 repeats the application evaluation under sequence-cluster controls. Candidate Hit@10 and MRR use the top-50 retrieval files; candidate hit@50 and strict typed path use the fixed 20-paper Agent evidence budget. The two views are reported together because raw paper hit at a 200-paper budget has a substantial chance baseline in this small, densely annotated graph.

Table 10: Budget-matched deterministic evidence evaluation on the 99-query SeqLit-DAG split. GO bridge recall measures expected GO coverage; strict path requires a typed candidate-to-GO-to-expected-PMID route.

Method	Candidate hit	Candidate recall	GO bridge recall	Citation hit	Strict path	Citation precision
Random	0.061	0.027	0.045	0.000	0.000	0.000
k-mer Jaccard	0.313	0.127	0.248	0.152	0.152	0.017
BLASTP	0.242	0.124	0.194	0.222	0.222	0.026
ESM-2 mean	0.364	0.181	0.301	0.202	0.202	0.021
ProtT5 mean	**0.465**	**0.225**	**0.368**	0.253	0.253	0.027
ProtT5+BLAST RRF	0.444	0.220	0.353	**0.263**	**0.263**	**0.027**

Table 11: Cluster-held-out SeqLit-DAG retrieval and evidence paths on 100 queries per split. Identity-30 is a deliberately cluster-stratified stress test. Strict path requires an explicit retrieved candidate–GO–expected-PMID edge within 20 papers.

Split	Route	Candidate Hit@10	Candidate MRR	Candidate hit@50	Strict path@20
UniRef50	Random	0.040	0.022	0.160	0.040
UniRef50	3-mer Jaccard	0.240	0.165	0.410	0.220
UniRef50	BLASTP	0.340	0.257	0.340	0.340
UniRef50	ProtT5 mean	0.430	0.317	0.620	0.390
UniRef50	ProtT5+BLAST RRF	0.440	0.322	0.620	0.420
Identity-30	Random	0.070	0.020	0.190	0.020
Identity-30	3-mer Jaccard	0.130	0.062	0.270	0.100
Identity-30	BLASTP	0.150	0.108	0.160	0.160
Identity-30	ProtT5 mean	0.220	0.154	0.390	0.190
Identity-30	ProtT5+BLAST RRF	0.200	0.126	0.370	0.180

ProtT5 minus BLASTP candidate MRR is $+0.061$ (95% paired bootstrap interval $[0.004, 0.120]$) on UniRef50 and $+0.047$ ($[0.008, 0.091]$) on identity-30. Because relevance is shared low-frequency GO plus index-side GOA evidence, these deltas support a complementary function-linked candidate route rather than superiority as an aligner. RRF is mixed: its UniRef50 Hit@10 delta over ProtT5 is $+0.010$ ($[-0.050, 0.070]$), whereas identity-30 MRR falls by -0.028 ($[-0.054, -0.007]$). We therefore retain fusion as a heuristic ablation.

We next tune Graph-IDF only on 33 development queries and freeze it on the remaining 66. The selected fused-route configuration uses the first five protein candidates, IDF power 1.5, three function claims, and five literature claims. Table 12 compares it with a rank-first selector at the same output budget. Unlike the earlier ProtT5-only selector ablation, the fused BLAST+ProtT5 route leaves little answer-F1 headroom for this heuristic: paired query bootstrap intervals for Graph-IDF minus the matched selector are $[-0.030, 0.026]$ for function and $[-0.003, 0.015]$ for literature. We therefore treat Graph-IDF as evidence compression and ordering, not as a statistically resolved retrieval improvement.

The cluster controls also isolate the DAG selector from the fixed generator. Table 13 reports literature F1 on each frozen 67-query test set. Graph-IDF gives a resolved literature-ordering gain on UniRef50 for both candidate routes, but neither identity-30 interval excludes zero. Moreover, on the UniRef50 fused route its function-F1 delta is -0.0560 ($[-0.1042, -0.0058]$). Typed graph selection is therefore task-specific evidence compression, not a generic retrieval or function-prediction improvement.

Finally, Qwen3.5-9B executes three structured questions for every frozen test pack: supported GO terms, supported PMIDs, and whether mechanism evidence is sufficient. We hold the generator, decoding, query split, prompt contract, candidate budget, and output budget fixed while changing only the evidence route. Table 14 therefore measures application-level evidence use rather than generator quality. The no-retrieval

Table 12: Frozen fused-route selector results on 66 accession-masked test queries. The retrieval oracle measures whether a correct identifier exists in the available pack and is not executable.

Method	Function F1	Function hit	Literature F1	Literature hit	ΔF1 function	ΔF1 literature
Rank-first, matched budget	0.0935	0.197	0.0837	0.258	–	–
Graph-IDF	**0.0938**	0.197	**0.0883**	0.258	+0.0003	+0.0045
Retrieval oracle	0.4054	0.485	0.1635	0.273	–	–

Table 13: Frozen cluster-control literature selection. Each row uses 33 development queries for selector tuning and 67 untouched test queries. The oracle is a pack-coverage ceiling, not an executable method.

Split / route	Rank-first F1	Graph-IDF F1	Retrieval oracle F1	Graph minus rank-first (95% CI)
UniRef50 / ProtT5	0.0812	0.0990	0.2126	$+0.0178$ $[0.0030, 0.0322]$
UniRef50 / fused	0.0724	0.1006	0.2174	$+0.0283$ $[0.0114, 0.0450]$
Identity-30 / ProtT5	0.0401	0.0498	0.0914	$+0.0096$ $[-0.0031, 0.0240]$
Identity-30 / fused	0.0334	0.0338	0.0926	$+0.0004$ $[-0.0138, 0.0137]$

condition invokes the system's no-evidence abstention policy; its zero answer F1 is a safety fallback, not a test of parametric biological knowledge.

The live R2R 3.6.6 control contains 14,071 documents and uses the frozen Qwen3-Embedding artifact above. At 50 retrieved chunks it returns 22–50 unique protein accessions per query but places no expected candidate, GO term, or PMID into the fixed five-candidate prompt. Sequence-vector evidence therefore raises function/literature F1 by 0.072/0.075 relative to R2R, with paired 95% intervals $[0.035, 0.116]$/$[0.039, 0.115]$; its GO/PMID prompt-recall gains of 0.187/0.128 have intervals $[0.106, 0.278]$/$[0.069, 0.195]$. Integrity checks localize this result to sequence representation rather than collection failure: five natural-language accession/title queries retrieve the target at rank one, whereas 20 exact indexed-protein sequence queries obtain Hit@1/10/50 of 0.05/0.05/0.10. Expanding R2R to 200 chunks recovers a relevant accession for 8/66 held-out queries but still supplies no expected label among the first five candidates. These diagnostics are not additional Agent conditions.

Relative to vector-only evidence, the combined route raises function/literature F1 by 0.015/0.009 and prompt recall by 0.015/0.005; the paired 95% intervals are $[0.000, 0.038]$/$[-0.003, 0.023]$ for F1 and $[0.000, 0.045]$/$[-0.004, 0.018]$ for prompt recall. DAG compression leaves prompt recall unchanged, as required by the matched evidence budget, but raises GO evidence-selection F1 from 0.933 to 1.000 (paired delta 0.067, 95% CI $[0.035, 0.105]$). Its function/literature answer-F1 deltas of 0.006/0.005 have intervals $[-0.026, 0.038]$/$[-0.003, 0.015]$ and are therefore directional rather than statistically resolved. Every evidence-bearing output has entailed citations and zero out-of-pack GO/PMID identifiers, and all mechanism prompts abstain. In particular, Qwen3.5 copies the non-gold R2R evidence without adding a gold or out-of-pack identifier, separating retrieval failure from generation-grounding failure.

Qwen3.5 generation uses BF16 and peaks at 17.88 GiB across all routes, so the complete experiment fits a 32 GB GPU. Resident-model generation averages 2.72 s per R2R answer and 2.83–3.00 s per sequence-evidence answer, with route-level P95 values of 4.69–6.28 s. The installed environment uses the model's PyTorch fallback rather than optional fused linear-attention kernels, so these are reproducibility measurements rather than optimized serving claims. They exclude retrieval, embedding, R2R network calls, and model loading. On the identical DAG packs, Qwen2.5-7B-Instruct obtains function/literature F1 of 0.094/0.090, evidence-selection F1 of 0.990/0.988, citation entailment 1.000, zero out-of-pack identifiers, and complete mechanism abstention. The close same-pack result supports generator robustness; model choice is not treated as a contribution.

Table 14: Fixed-Qwen3.5 Agent route ablation on 66 held-out queries. The R2R row is a live generic semantic text-RAG control over the same held-out index. Prompt recall measures whether expected labels reach the executable prompt; entailment checks every emitted citation against the supplied evidence.

Evidence route	Function F1	Literature F1	GO prompt recall	PMID prompt recall	GO selection	PMID selection	Entailment	Hallucination	Mechanism abstention
No retrieved evidence	0.000	0.000	0.000	0.000	–	–	–	0.000	1.000
Ordinary text RAG (R2R)	0.000	0.000	0.000	0.000	0.865	1.000	1.000	0.000	1.000
Sequence vector	0.072	0.075	0.187	0.128	0.944	1.000	1.000	0.000	1.000
Combined BLAST+vector	0.087	0.084	0.202	0.133	0.933	1.000	1.000	0.000	1.000
Combined BLAST+vector+DAG	**0.094**	**0.088**	0.202	0.133	**1.000**	1.000	1.000	0.000	1.000

The low answer F1 relative to citation reliability identifies the next bottleneck: increasing retrieval and compact-selector coverage without admitting a noisy paper pool. Expanding the paper budget from 20 to 200 doubles the test literature-oracle hit from 0.258 to 0.515, but does not improve the current selector, demonstrating that unfiltered graph expansion raises both ceiling and noise. The live R2R route takes 0.833 s/query on average (P95 1.432 s), including server-side Qwen3 query embedding and pgvector lookup. It is an application baseline rather than a speed claim; optimized text serving and concurrency are not evaluated.

7 Discussion

7.1 Why Vector Retrieval Matters Even When BLAST Is Stronger

BLAST is stronger for alignment-grounded verification and should remain the baseline for exact sequence similarity. However, agent systems need more than exact sequence matching. They need a unified evidence interface that can retrieve text, sequences, annotations, and graph paths, then package them into prompt-ready citations. Vector retrieval provides this shared substrate. The held-out ESM-2 and ProtT5 results further suggest that BioRAG should not force all partitions through one embedding model: specialized protein encoders can be used where they are stronger, while OmniGene remains useful for unified biological-language and mixed-modality contexts. In the cluster controls, ProtT5's advantage concerns function-linked candidates and literature paths, while BLAST retains alignment semantics. The right comparison is not vector versus BLAST as mutually exclusive tools, but vector as coarse candidate generation and BLAST as fine verification.

7.2 Lookup and Verified Retrieval Modes

The latency results motivate two operating modes. Lookup mode returns vector-only context after query embedding and is useful for rapid local evidence preview. Verified mode runs BLAST and DRAG expansion to attach biologically grounded evidence. This design fits biomedical agents: the user can receive provisional context from a unified local index, while the system appends verified evidence when the slower routes finish. The DNA matrix now reports query embedding plus lookup latency for DNABERT-2; the older Chroma stress-test measurements remain separately labeled and do not include model cold start. On the 2k cluster controls, resident ProtT5 BF16 query embedding plus exact NumPy lookup takes 19.04 ms/query for UniRef50 and 17.35 ms/query for identity-30 on an RTX 4080, with 2.59 GiB peak allocated memory. These timings exclude model load, graph expansion, generation, and network calls.

7.3 DRAG as a Biological Representation Probe

The graph results suggest that sequence-vector neighborhoods may contain biological structure, not just retrieval convenience. The current main-text claim is intentionally narrow: DRAG provides an inspectable evidence package for agents, and its sequence communities are promising exploratory objects. Initial controls sharpen this boundary. The analyzed graph views are already parent-collapsed, and degree-preserving null graphs do not show comparable label-purity signals, which argues against pure topology as the explanation. However, k-mer/Jaccard graphs also recover strong family-level communities, so part of the observed DRAG signal is explained by ordinary sequence similarity. Stronger biological-discovery claims require larger parent-collapsed graphs and domain, GO/pathway, and literature case studies against these controls.

8 Limitations

BioRAG-Standard v0 is still a first version. The main 100-query sequence test uses fragments or perturbations from indexed parent sequences and should therefore be interpreted as a local workflow stress test rather than a held-out biological homology benchmark. The SeqLit-DAG application adds a leakage-controlled 99-query accession-masked path task and two 100-query cluster controls, but all are sampled from the same 2k human-protein graph. The observed UniRef50 sample is mostly singleton clusters, while identity-30 deliberately oversamples non-singleton components and is therefore a stress test rather than a prevalence estimate. Neither controls Pfam clan, taxonomy, time, or remote homology. Relevance is defined by low-frequency shared GO labels and index-side GOA papers; those papers are traceable function-linked evidence, not automatically direct experimental support for the held-out protein. The exact-identifier, two-line Agent task is auditable but substantially narrower than free-form biomedical QA, which still requires expert review and broader datasets. The mixed partition should also be expanded with abstracts, UniProt function text, pathway descriptions, and sequence snippets.

The current vector reranker is lightweight and heuristic, and candidate-subset BLAST reranking is limited by whether the vector candidate pool contains the correct or biologically equivalent parent sequence. A preliminary graph-expanded retrieval ablation used 10k view graphs rather than a full 100k or complete sequence graph; it added graph candidates but did not improve recall beyond the vector seed pool, so it is treated as a coverage-limited result rather than a central claim. FAISS GPU lookup is now measured on an RTX 4080 SUPER for the controlled20k DNA collection, but this remains a lookup-only microbenchmark rather than a production throughput result; larger-scale serving and concurrency are still pending. Functional and literature enrichment coverage depends on local UniProt/NCBI/HGNC mappings; Pfam or InterPro domain enrichment and title-level literature case studies would strengthen biological interpretation.

The core protein embedding comparison is complete for the current held-out controls: ProtT5 mean pooling is the selected protein-partition default, ESM-2 mean pooling is a lighter alternative, and OmniGene remains the unified mixed-modality backbone. We also added ESM-2 and ProtT5 mean-vs-last pooling ablations and Nucleotide Transformer, DNABERT-S, DNABERT-2, and DNA-ESM2-style DNA controls. DNABERT-2 mean pooling is the strongest tested DNA dense condition, but it remains an auxiliary candidate layer rather than a replacement for BLASTN. Larger held-out DNA/family splits and full-scale DNA vector indexes remain useful follow-ups, while original Gemma4 MoE controls are deferred until after the first review. The tokenizer activation audit means that current experiments separate the BioRAG system and sequence-CPT checkpoint from neither a corrected modality-aware biological tokenizer nor its required retraining; no vocabulary-gain claim is made. The BLAST side of the Swiss-Prot scale curve and the controlled100k/300k ProtT5 vector/candidate-BLAST points are now measured, but a full-scale dense model comparison, FAISS/Milvus-style optimized serving, and candidate-subset BLAST sweeps are still needed before making full-corpus model or speed claims.

The biological-meaning analysis should also be interpreted carefully. Community purity, GO/Reactome enrichment, and shared PubMed IDs show that DRAG neighborhoods are non-random and inspectable, but they do not prove causal mechanisms or validate novel wet-lab hypotheses. The exploratory token graph likewise does not improve the primary ProtT5 ranking: naive graph fusion is harmful, and development-selected tail replacement recovers only one additional top-100 test candidate without changing paper retrieval. It is retained as an explanatory annotation path, not as a retrieval contribution. Graph-IDF improves frozen literature selection on the UniRef50 control, but the identity-30 result is unresolved and one fused-route function score is significantly worse; no universal DAG ranking gain is claimed. The generated Agent experiment checks structured identifier correctness and typed-edge citation entailment; it does not judge narrative biological reasoning. A 30-case, five-route assessor-blinded free-form package has been frozen, but its domain-expert ratings are pending and therefore contribute no biological-correctness claim. Its automatic integrity audit is likewise not a substitute for expert assessment. The R2R comparison uses one Qwen3-Embedding-0.6B Q8_0 artifact and one human-protein task; it demonstrates the need for modality routing in this deployment, not a universal failure of general-purpose text embedders. The appropriate use is hypothesis generation, evidence organization, and agent grounding. Biological claims that affect experimental design, diagnosis, treatment, or safety-critical decisions should still be verified by domain experts and specialized tools.

9 Reproducibility and Use Scope

The implementation is local-first and will be released at `https://github.com/maris205/biorag`. The corpus export lives under `data/biorag_standard_v0`; the SeqLit-DAG resources live under `data/seq_lit_dag_*`; retrieval and graph reports are stored under `reports/`; and the main evaluation scripts are in `scripts/`. The main BioRAG sequence-vector experiments use the full OmniGene-4 CPT merged model in BF16, not the GGUF or 4-bit quantized variants, while held-out protein partitions use ESM-2 or ProtT5 BF16 embeddings and the DNA control uses DNABERT-2 mean pooling. The generated Agent ablation uses Qwen3.5-9B BF16 as a separate fixed instruction model and Qwen2.5-7B-Instruct as a robustness control; neither uses the biological embedding backbone as the answer generator. The vector store is Chroma for the POC because it supports simple CRUD and collection management, while FAISS CPU/GPU provide lookup-speed references. The ordinary text-RAG control uses R2R 3.6.6, PostgreSQL 14.24, pgvector 0.8.1, Ollama 0.33.2, and a frozen Qwen3-Embedding-0.6B Q8_0 artifact; R2R remains an application integration rather than the scientific sequence index. BLAST runs against local Swiss-Prot and Ensembl cDNA databases.

The main reported commands are documented in `docs/REPRODUCIBILITY.md`, the evaluation notes, and the research matrix, including vector candidate-budget sweeps, vector-to-BLAST reranking, lookup/verified latency decomposition, DRAG construction, cluster-held-out SeqLit-DAG export, paired bootstrap analysis, Graph-IDF selection, structured Agent scoring, and the frozen free-form review protocol. The public repository tracks source code, task definitions, paper source, and compact result summaries; large local data files, model checkpoints, Chroma indexes, BLAST databases, PubMed text caches, assessor keys, and per-query JSON traces are regenerated from scripts rather than versioned. The Chroma and FAISS lookup-only tables exclude query embedding; the cluster-control ProtT5 timings report query embedding and lookup separately while excluding model load. R2R timing includes its server-side Qwen3 query embedding and pgvector lookup, whereas all retrieval timings exclude LLM generation. Agent generation latency is reported separately with its model-load boundary.

BioRAG-DRAG is intended for research and evidence organization, not autonomous biomedical decision-making. The system can help retrieve, rank, and package evidence for a human or an auditable agent workflow, but it should not be used as a standalone clinical or experimental authority. The design deliberately keeps BLAST, curated annotations, graph evidence, and literature traces visible so that downstream users can inspect which route supported each claim.

10 Conclusion

BioRAG-DRAG provides a unified local retrieval layer for biomedical agents by combining neural sequence-text retrieval, classical biological verification, and typed evidence graph expansion. Vector retrieval is useful as a coarse multimodal candidate layer, while BLAST remains essential for alignment-grounded verification. The SeqLit-DAG experiments add an executable application: a protein sequence can lead to compact function and literature identifiers that a local instruction model cites without leaving the retrieved evidence pack and explicitly declines to turn into unsupported mechanism claims. Observed-UniRef50 and identity-30 controls show that ProtT5 can complement local alignment for function-linked evidence candidates after stronger cluster exclusions, while simple rank fusion is not robust. Typed DAG selection improves literature ordering on the UniRef50 control but not uniformly across tasks or splits. In the fixed-generator accession-masked ablation, the live ordinary text-RAG control fails to supply expected raw-sequence evidence, sequence-aware retrieval gives a resolved gain, BLAST fusion directionally improves coverage, and DAG compression resolves GO evidence-selection errors without changing the candidate budget. Same-pack Qwen2.5 and Qwen3.5 results separate the latter effects from generator choice. The R2R deployment exposes the evidence layer through a production-style application boundary while keeping sequence retrieval and verification in BioRAG. Larger taxonomy-diverse sequence-to-literature benchmarks, stronger evidence selectors, free-form expert evaluation, and controlled biological discovery studies remain necessary before broader scientific claims.

A Candidate-BLAST Ablation

Table 15 reports the full vector-to-candidate-BLAST budget sweep. Each query is vector-searched once at the maximum budget; smaller budgets use prefixes of that candidate list, so this isolates candidate-pool quality and candidate-BLAST verification rather than per-budget vector lookup latency. This table supports the engineering hypothesis that vector search can provide candidate pools for later alignment verification, but it is not treated as a main retrieval condition.

Table 15: Vector-to-candidate-BLAST budget sweep on the 100-query sequence-window stress test.

Candidate budget	Exact Hit@10	Exact MRR	Biological Hit@10	Biological MRR	Candidate Bio Recall@N	Candidate BLAST latency
10	0.5000	0.3945	0.8283	0.8196	0.8283	172.0 ms
25	0.5400	0.4517	0.8485	0.8394	0.8485	133.1 ms
50	0.5700	0.4898	0.8687	0.8687	0.8788	131.6 ms
100	0.6100	0.5151	0.8990	0.8990	0.9293	138.1 ms
200	0.6500	0.5377	0.9293	0.9293	0.9596	146.2 ms

B Exploratory DRAG Biological-Structure Results

This appendix reports exploratory biological-structure analyses that are useful for hypothesis generation but are not used as main-text evidence for biological mechanism. The main text includes compact parent-collapse, k-mer/Jaccard, and degree-preserving null controls; the enrichment tables below should be read in light of those controls.

Table 16 adds a parent-level token graph over 2,000 Swiss-Prot proteins with experimental GOA/PubMed evidence. The intended frequency-derived protein BPE segmentation yields 15 FDR-significant token–GO pairs, whereas 100 length-stratified label permutations produce 0.12 on average (maximum 2). The strongest observed edge is `HRD` to protein serine/threonine kinase activity. However, overlapping fixed 3-mers recover 57 significant pairs and more associated GO terms. This supports the narrow claim that some candidate sequence tokens coincide with biologically informative local patterns, but it provides no evidence that BPE is superior to ordinary k-mers, that the token itself is a curated motif, or that the inactive added entries acquired these semantics through CPT.

Table 16: Exploratory parent-level protein token–GO association controls. Significant pairs use BH-FDR $q \leq 0.05$ and at least three supporting proteins. The null permutes complete GO label sets within protein-length strata.

Tokenization	Token positions	Eligible units	Significant pairs	Significant units	Null mean/max
Intended protein BPE	402,404	4,342	15	13	0.12/2
Current runtime tokens	548,360	1,465	15	14	0.09/2
Overlapping fixed 3-mer	1,055,490	7,915	57	53	0.06/2

The exported exploratory extension uses exact `observed_in_sequence` edges from token nodes to proteins and computed `statistically_associated_with` edges from token nodes to GO terms. The latter are explicitly marked as exploratory statistical associations; curated GOA evidence and PubMed paths remain separate. A paper-grade token-semantics claim requires PROSITE/Pfam coordinate overlap, adjacent token-composition ablations, and UniRef50 family-held-out evaluation.

Table 17 asks whether that token graph improves held-out retrieval. Direct BPE overlap and BPE-to-GO expansion are both below the fixed 3-mer and ProtT5 controls. Reciprocal-rank fusion reduces ProtT5 Recall@100 from 0.2737 to 0.2334 and MRR from 0.2094 to 0.1180. A development-selected tail route keeps the first 75 ProtT5 candidates and fills the remainder from direct BPE retrieval. It preserves Hit@10/50 and recovers one additional test query at top 100, but the paired Hit@100 delta of 0.0152 has a 95% interval of $[0.0000, 0.0455]$, the Recall@100 interval crosses zero, and paper retrieval is unchanged. Adding GO expansion produces the same selected tail ranking. These results do not support using the current token graph as a primary retriever.

Table 17: Exploratory sequence-token retrieval on the frozen 66-query parent-held-out test split. Every token–GO edge is learned from index proteins only. The split is not family-held-out.

Route	Hit@10	Hit@50	Hit@100	Recall@100	MRR	Paper hit
BPE BM25	0.0909	0.1970	0.3788	0.1092	0.0588	0.2576
BPE+GO graph	0.0758	0.1818	0.3333	0.0877	0.0558	0.2273
Fixed 3-mer BM25	0.0909	0.2727	0.4091	0.1540	0.0806	0.3636
ProtT5 vector	0.2879	0.5152	0.5758	0.2737	0.2094	0.5455
ProtT5+BPE tail	0.2879	0.5152	0.5909	0.2770	0.2095	0.5455

Table 18: DRAG graph structure on 10k sequence-window graph views.

Modality	Graph	Nodes	Edges	Components	Communities	Modularity	Key biological signal
DNA/cDNA	vector-only	579	19,344	2	9	0.1741	IGKV community: 50/66, enrichment $\times$6.178
DNA/cDNA	BLAST-only	579	995	219	220	0.9383	compact IGHV/IGKV/IGLV alignment modules
DNA/cDNA	hybrid	579	19,592	2	7	0.1739	IGKV 51/60, enrichment $\times$6.9317; IG_J and IG_D modules
Protein	vector-only	2,623	31,794	2	16	0.3179	pgl 51/306 $\times$4.4158; yfbR 42/226 $\times$11.6062
Protein	BLAST-only	2,623	7,572	421	447	0.9721	compact but fragmented alignment components
Protein	hybrid	2,623	35,549	1	12	0.3779	pgl 71/519 $\times$3.6245; yfbR 42/63 $\times$41.6349

Table 19: Vector-neighborhood enrichment against random neighbors.

Target	Match	Vector Hit@10	Random Hit@10	Vector P@10	Random P@10
Protein windows	`GN=` / gene symbol	0.4300	0.0200	0.1710	0.0020
DNA/cDNA windows	gene ID	0.6900	0.0600	0.2460	0.0060
DNA/cDNA windows	gene symbol	0.7188	0.0625	0.2563	0.0063
DNA/cDNA windows	any identity	0.7100	0.0600	0.2570	0.0060

Table 20: GO and Reactome enrichment of DRAG graph communities.

Graph	GO annotated nodes	Reactome annotated nodes	Top GO signal	Top Reactome signal
DNA/cDNA vector-only	155	68	proton transmembrane transport, 11/52, $q = 0.000113$	Metabolism, 12/26, $q = 0.000015$
DNA/cDNA BLAST-only	155	68	immunoglobulin mediated immune response, 11/11, $q = 0.000019$	no mapped top term
DNA/cDNA hybrid	155	68	proton transmembrane transport, 11/50, $q = 0.000088$	Metabolism, 12/26, $q = 0.000015$
Protein vector-only	63	213	intracellular protein localization, 6/8, $q = 0.000516$	metabolism of steroid hormones, 6/26, $q = 0.000452$
Protein BLAST-only	63	213	postsynaptic membrane, 5/5, $q = 0.000042$	GPCR downstream signalling, 9/25, $q = 0.000126$
Protein hybrid	63	213	focal adhesion, 5/8, $q = 0.000539$	TP53 regulates metabolic genes, 7/42, $q = 0.000223$

Table 21: Shared PubMed evidence in DRAG graph communities.

Graph	NCBI-mapped nodes	PubMed nodes	Unique PMIDs	Shared-PMID communities	Top shared PMID signal
DNA/cDNA vector-only	197	196	1,092	7	PMID:19050702, 12/63, $q = 0.000314$
DNA/cDNA BLAST-only	197	196	1,092	20	PMID:8490662, 9/24, $q = 0.000596$
DNA/cDNA hybrid	197	196	1,092	6	PMID:20301403, 11/59, $q = 0.000312$
Protein vector-only	63	63	5,856	4	PMID:11697890, 6/8, $q = 0.000146$
Protein BLAST-only	63	63	5,856	13	PMID:25402006, 4/4, $q = 0.000682$
Protein hybrid	63	63	5,856	4	PMID:27432908, 8/8, $q = 0.000159$

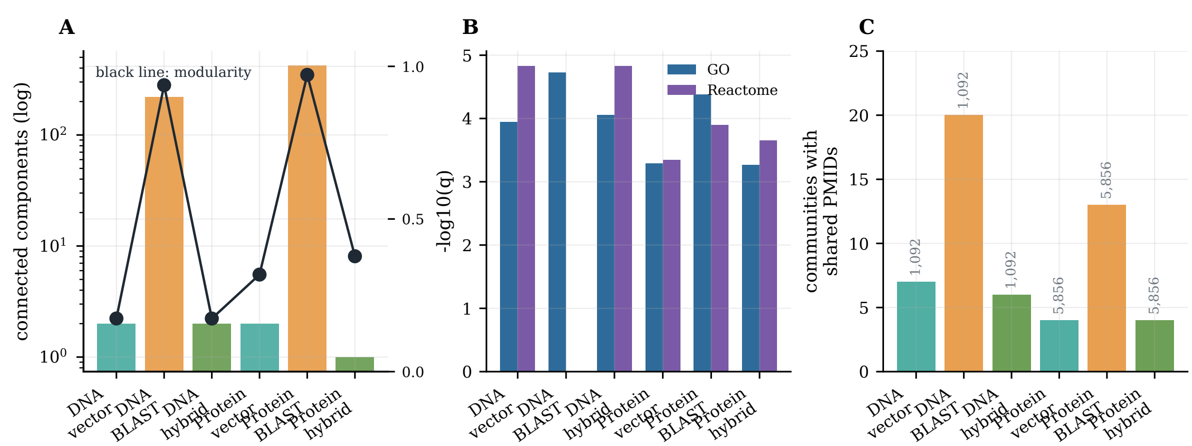

Figure 7: Exploratory DRAG biological-structure analysis. Vector, BLAST, and hybrid graphs expose different graph connectivity, functional enrichment, and literature-sharing patterns. These signals support hypothesis generation and should be interpreted with the parent-collapse, k-mer/Jaccard, and degree-preserving null controls in Table 8.

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